BAXI 2213: Knowledge Based System

Structure of KBS:

1. KBS Structure
2. Reasoning

KBS Structure

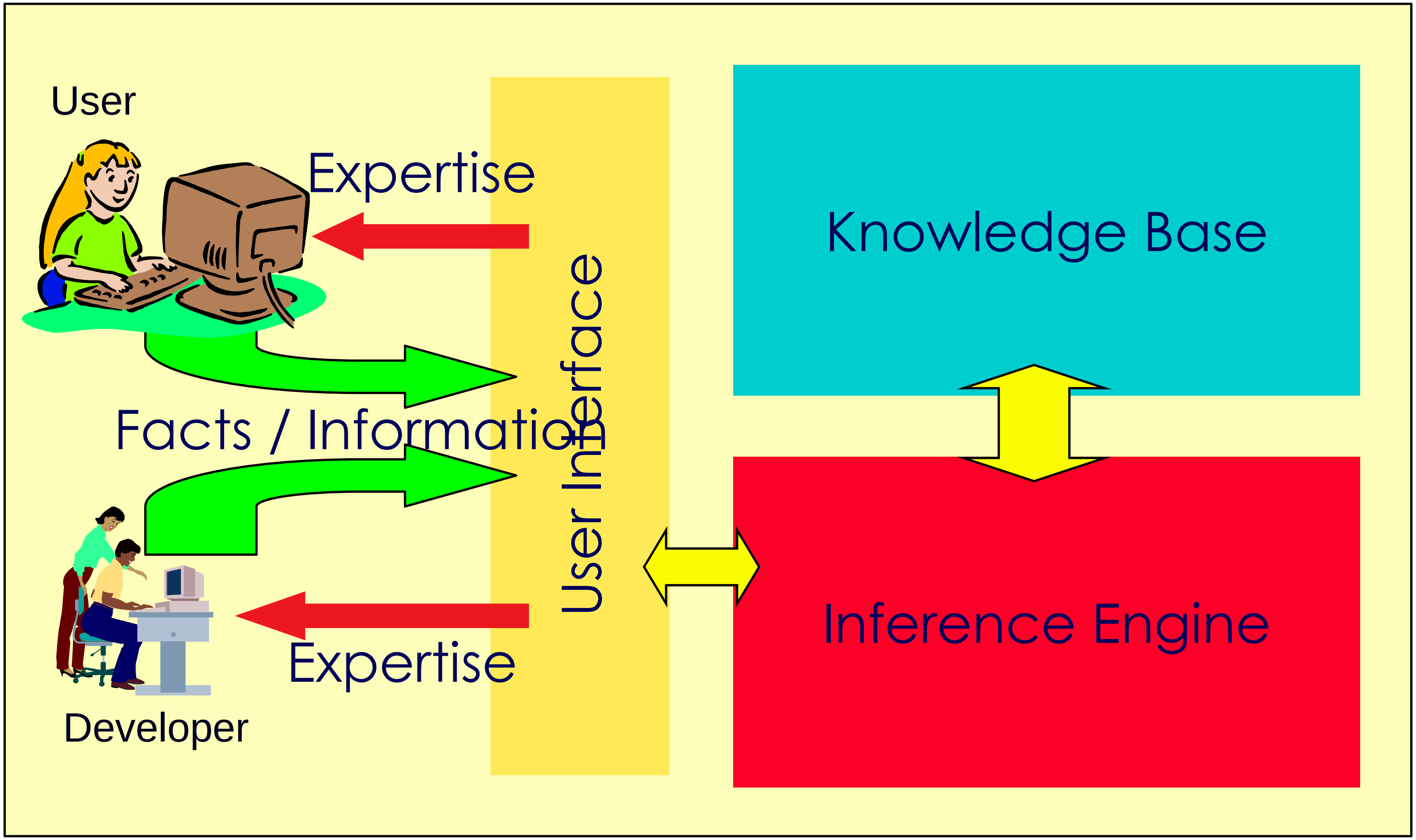
Structure of a KBS: Viewpoints

- There are 3 principle roles for those who work with KBS:

End User	• Doctor, factory manager, person of the street, etc.
Knowledge Engineer	• The person who designs, classifies, organizes, builds a representation of ... the knowledge)
Builder	• Typically a programmer

- Often, the last **2 roles** are held by the same person (less so on big projects).

The End User's View



REASONING

Inference Engine

- The inference engine controls the **reasoning** involved when the system is run.
- It has its own mechanism for interpreting the stored knowledge (in the appropriate form), and for sequencing the steps involved in reaching conclusions.
- ***Inference*** here means any of the methods by which the system reaches conclusions.
- The inference process done by human is called *reasoning*.

Reasoning

- The proses of working with knowledge, facts and problem solving strategies to draw conclusions.

- Understanding how humans reason, how they work information on a given problem coupled with their general knowledge of the domain, provides insight into how to guide knowledge processing in an expert system.

- Some types of reasoning: deductive, inductive, analogical and common-sense.

Deductive Reasoning

- Deduce (infer/ assume/ reason) new information from logically related known information.
- Use problem facts and related general knowledge in a form of rules or implications.
- Process begin with comparing the facts with a set of implications to conclude new facts.

Implication : I will get wet if I am standing in the rain
Fact : I am standing in the rain
Conclusion : I will get wet

Deductive Reasoning

Fact : All animals breathe oxygen.
Fact : All cats are animals.
Conclusion : All cats breathe oxygen.

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Inductive Reasoning

- Arrive at a general conclusion from a limited set of facts by the process of generalization.
- Through inductive reasoning, we form a generalization which we believe applies to all cases of a certain type, on the basis of a limited number of cases.

Fact : Person A says Data Structure subject is difficult.
Fact : Person B says Data Structure subject is difficult.
Conclusion: You believe Data Structure subject is also difficult.

Inductive Reasoning

You have a very good friend circle. Therefore, you are very good.

Prolog is easy, Java is easy. So, programming subjects are easy.

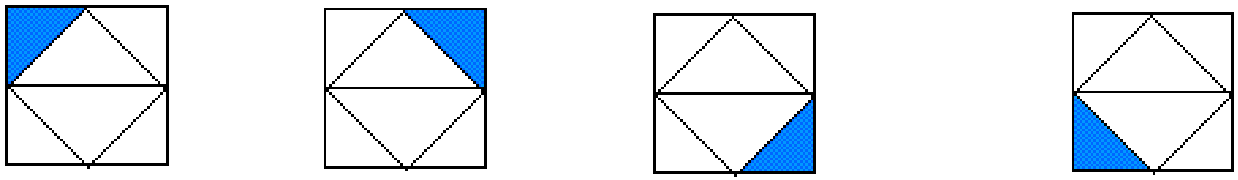
Guess the missing numbers:

10 20 30 40 …

1 2 4 7 11 16 ...

Inductive Reasoning

Look at the patterns below:



What is the next pattern you guess?

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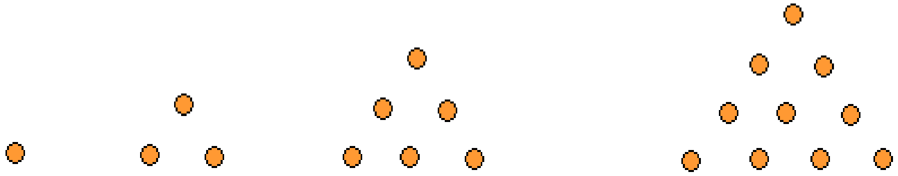
What is the next pattern you guess?



Analogical Reasoning

- Humans form a mental model for some concept through their experiences.
- Use this model through analogical reasoning to help in understanding some situation or object.
- Draw analogies between two, looking for similarities and differences to guide reasoning.

Analogical Reasoning

- Is an apple more like an orange or a banana?"
- Although everyone probably knows the differences between these three types of fruit, just how would we go about making a valid analytical comparison of them?
- Analogical reasoning seeks to identify specific sets of similar and dissimilar characteristics, in search of some unique combination of characteristics that can then be used to define distinctive properties of each set.

Analogical Reasoning

	Apple	Orange	Banana
Edible seeds	NO	NO	YES
Shape	Round	Round	Oblong
Edible skin	YES	NO	NO
Peelable without a knife	NO	YES	YES
Citrus family	NO	YES	NO
Color when ripe	Red, Orange, Yellow, Green	Orange only	Yellow only

- In making the final comparison, we then attempt to determine in which of these categories apples and oranges are most alike (i.e., similar).
- Apples and oranges are similar only in that they are both generally round in shape and their seeds are not edible (eatable).
- In all other categories of our comparison, apples and oranges are dissimilar.

Common Sense Reasoning

- Through experience, humans learn to solve problems efficiently.
- They use their common sense to quickly derive a solution.
- Common sense reasoning depend on good judgments than on exact logic.

Common Sense Reasoning

"Do you have one of those phones you can talk to people on?"

Common Sense Reasoning

Method	*Description*
Deductive reasoning	Move from a general principle to a specific inference. A general principle is composed of two or more premises.
Inductive reasoning	Move from some established facts to draw general conclusions.
Analogical reasoning	Derive an answer to a question by known analogy. Analogical reasoning is a verbalization of an internalized learning process (see Owen, 1990). It involves use of similar past experiences.
Formal reasoning	Syntactically manipulate a data structure to deduce new facts, following prescribed rules of inferences (e.g., predicate calculus).
Procedural (numeric) reasoning	Use mathematical models or simulation (such as model-based reasoning, qualitative reasoning, and temporal reasoning, or the ability to reason about the time relationships between events).
Metalevel reasoning	Have knowledge about what is known (e.g., about the importance and relevance of certain facts and rules).